

Информация

Докладчик

- надиа эззакат
- Студент физмат
- Российский университет дружбы народов

Цели и задачи работы

Целью данной лабораторной работы является ознакомление с работа с структура документа Latex

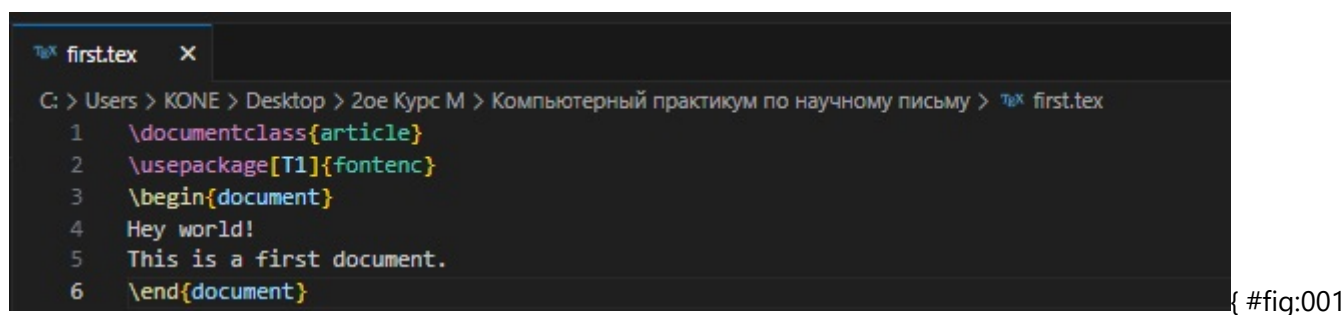
Задачи: Рассмотреть и работа с документа структура TexLive:

1. Try adding text to your first document, typesetting and seeing the changes in your PDF.
2. Make some different paragraphs and add variable spaces.
3. Explore how your editor works; click on your source and find how to go to the same line in your PDF.
4. Try adding some hard spaces and see how they influence line-breaking.

Теоретическое введение

```
\documentclass{article}
\usepackage[T1]{fontenc}
\begin{document}
Hey world!
This is a first document.
\end{document}
```

(см. Рис. [-@fig:001]).



width=100% }

Running LaTeX

(см. Рис. [-@fig:002]).

```
Windows PowerShell  
Copyright (C) Microsoft Corporation. Tous droits réservés.  
  
Installez la dernière version de PowerShell pour de nouvelles fonctionnalités et améliorations ! https://aka.ms/PSWindows  
  
PS C:\Users\KONE\Desktop>2oe Курс М\Компьютерный практикум по научному письму> pdflatex first.tex  
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdflatex)  
restricted \write18 enabled.  
entering extended mode  
(./first.tex  
LaTeX2e <2025-06-01> patch level 1  
L3 programming layer <2025-09-02>  
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls  
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class  
(c:/texlive/2025/texmf-dist/tex/latex/base/size12.clo))  
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)  
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdfptex.def)  
No file first.aux.  
  
[!{c:/texlive/2025/texmf-var/fonts/map/pdfptex/updmap/pdfptex.map}] (./first.aux)  
)  
kpathsea: Running mktexpk --mfmode / --bdpi 600 --mag 1+0/600 --dpi 600 ecrm1000  
  
The command name is C:\texlive\2025\bin\windows\mktexpk  
mf-nowin.exe --progname=mf --base=mf \mode:=ljfour; \mag:=1+0/600; nonstopmode; input ecrm1000;  
This is METAFONT, Version 2.71828182 (TeX Live 2025) (preloaded base=mf)  
  
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/ecrm1000.mf  
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exbase.mf)  
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/ecrm.mf  
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exroman.mf  
Ok (c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exaccess.mf  
Ok) (c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/expseudo.mf  
Ok)(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exrwest.mf  
ok [ecrl] [ecsl] [ecpl] [ecfl] [eccl] [ecdl] [ecel] [ecol] [ecal] [ecal]
```

```
{ #fig:002 width=100% }
```

(см. Рис. [-@fig:003]).

```
[183] [184] [185] [186] [187] [188]]
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exbraces.mf
ok [94] [126] [23] [40] [41] [60] [124] [62] [91] [93] [92] [123] [125]
[95] [127] [32])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/expunct.mf
ok [14] [15] [19] [20] [13] [18] [33] [39] [42] [43] [44] [46] [47] [58]
[59] [61] [96] [189] [17] [45] [16] [21] [22])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exaccent.mf
ok [0] [1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] [12])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exsign.mf
ok [24] [34] [35] [36] [37] [64] [191] [159])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exrlig.mf
ok [25] [26] [28] [27] [29] [30] [31])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exromp.mf
ok [38] [63] [190])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exrulett.mf
ok [65] [66] [67] [68] [69] [70] [71] [72] [73] [74] [75] [76] [77] [78]
[79] [80] [81] [82] [83] [84] [85] [86] [87] [88] [89] [90])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exrlllett.mf
ok [97] [98] [99] [100] [101] [102] [103] [104] [105] [106] [107] [108]
[109] [110] [111] [112] [113] [114] [115] [116] [117] [118] [119] [120]
[121] [122]) (c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exrdigit.mf
ok [48] [49] [50] [51] [52] [53] [54] [55] [56] [57])
(c:/texlive/2025/texmf-dist/fonts/source/jknappen/ec/exrligtb.mf
ok) ) )
(some charht values had to be adjusted by as much as 0.07143pt)
Font metrics written on ecrml200.tfm.
Output written on ecrml200.600gf (256 characters, 60320 bytes).
Transcript written on ecrml200.log.
gftopk.exe ecrml200.600gf ecrml200.600pk
<c:/Users/KONE/.texlive/2025/texmf-var/fonts/pk/ljfour/jknappen/ec/e
crml200.600pk>
Output written on first.pdf (1 page, 10182 bytes).
Transcript written on first.log.
PS C:\Users\KONE\Desktop\2oe Курс M\Компьютерный практикум по научному письму
```

```
{ #fig:002 width=100% }
```

(см. Рис. [-@fig:004]).

🖥️ > Bureau > Zoe Курс M > Компьютерный практикум по научному письму >

📁 📄 🔄 🗑️ ⬆️ ⬆️ Trier ▾ ≡ Afficher ▾ ⋮

📁 Brouillon

📁 lab01

📁 lab02

📄 first.aux

📄 first.log

📄 first.pdf

📄 first.tex

📄 Practical-scientific-writing.pdf

{ #fig:002 width=100% }

(см. Рис. [-@fig:005]).

```
\documentclass{article}
\usepackage[T1]{fontenc}
```

```
\begin{document}
Hey world!
```

```
This is a first document.
\end{document}
```

Save the file and run:

```
pdflatex first.tex
```

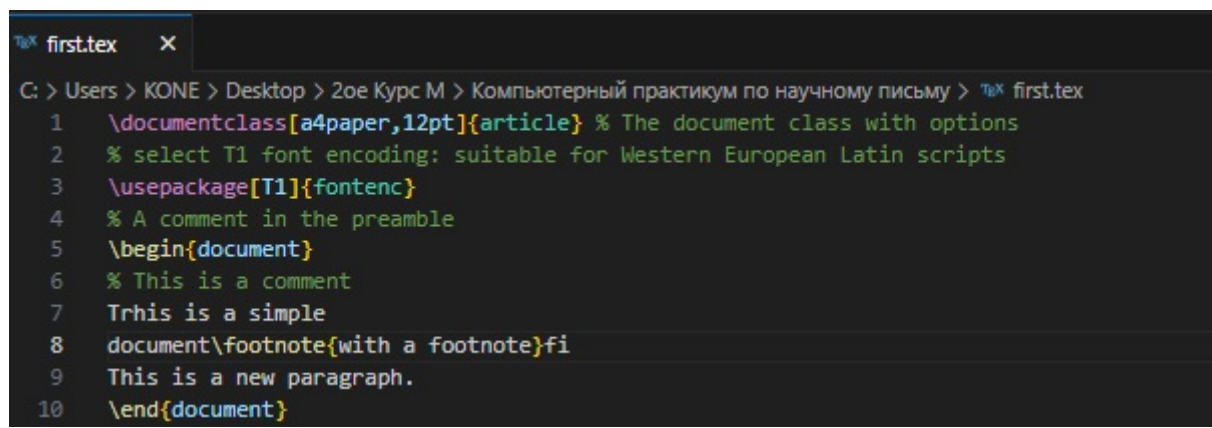
Hey world! This is a first document.

{ #fig:003 width=100% }

Документ TeXlive с нижним колонтитулом

```
\documentclass[a4paper,12pt]{article} % The document class with options
% select T1 font encoding: suitable for Western European Latin scripts
\usepackage[T1]{fontenc}
% A comment in the preamble
\begin{document}
% This is a comment
Trhis is a simple
document\footnote{with a footnote}fi
This is a new paragraph.
\end{document}
```

(см. Рис. [-@fig:006]).



```

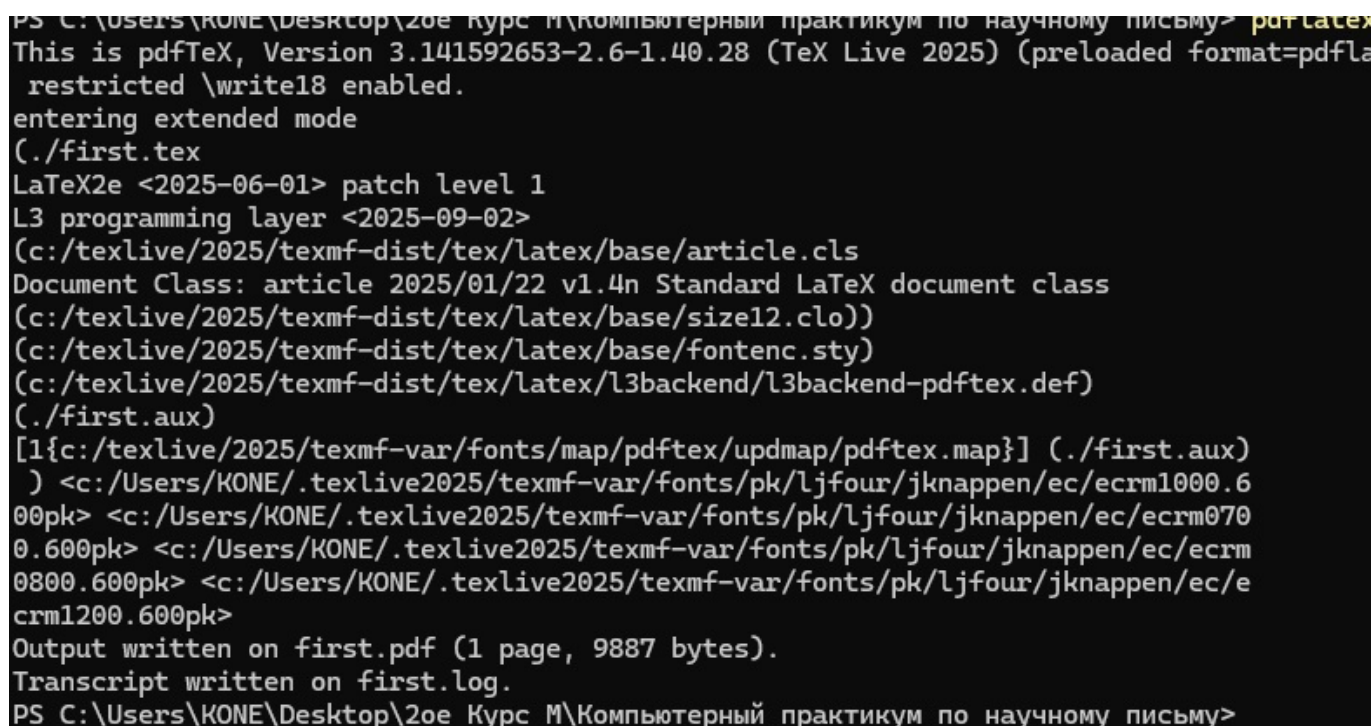
1  \documentclass[a4paper,12pt]{article} % The document class with options
2  % select T1 font encoding: suitable for Western European Latin scripts
3  \usepackage[T1]{fontenc}
4  % A comment in the preamble
5  \begin{document}
6  % This is a comment
7  This is a simple
8  document\footnote{with a footnote}fi
9  This is a new paragraph.
10 \end{document}

```

{ #fig:003

width=100% }

(см. Рис. [-@fig:007]).



```

PS C:\Users\KONE\Desktop\2oe Курс М\Компьютерный практикум по научному письму> pdflatex
This is pdfTeX, Version 3.141592653-2.6-1.40.28 (TeX Live 2025) (preloaded format=pdfla
restricted \write18 enabled.
entering extended mode
./first.tex
LaTeX2e <2025-06-01> patch level 1
L3 programming layer <2025-09-02>
(c:/texlive/2025/texmf-dist/tex/latex/base/article.cls
Document Class: article 2025/01/22 v1.4n Standard LaTeX document class
(c:/texlive/2025/texmf-dist/tex/latex/base/size12.clo))
(c:/texlive/2025/texmf-dist/tex/latex/base/fontenc.sty)
(c:/texlive/2025/texmf-dist/tex/latex/l3backend/l3backend-pdftex.def)
(./first.aux)
[1{c:/texlive/2025/texmf-var/fonts/map/pdftex/updmap/pdftex.map}] (./first.aux)
) <c:/Users/KONE/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecrm1000.6
00pk> <c:/Users/KONE/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecrm070
0.600pk> <c:/Users/KONE/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/ecrm
0800.600pk> <c:/Users/KONE/.texlive2025/texmf-var/fonts/pk/ljfour/jknappen/ec/e
crm1200.600pk>
Output written on first.pdf (1 page, 9887 bytes).
Transcript written on first.log.
PS C:\Users\KONE\Desktop\2oe Курс М\Компьютерный практикум по научному письму>

```

{ #fig:003 width=100% }

(см. Рис. [-@fig:008]).

This is a simple document¹. This is a new paragraph.

```
\documentclass[a4paper,12pt]{article} % T
% select T1 font encoding: suitable for W
\usepackage[T1]{fontenc}
% A comment in the preamble
\begin{document}
% This is a comment
This is a simple
document\footnote{with a footnote}.

This is a new paragraph.
\end{document}
```

You can see above that we've got two paragraphs: not
do that. Also notice that multiple spaces are treated as

¹ with a footnote

{ #fig:003 width=100% }

Выполнение лабораторной работы

Try adding text to your first document, typesetting and seeing the changes in your PDF.

(см. Рис. [-@fig:009]).


```

second.tex X
> Users > KONE > Desktop > 2oe Kypc M > Компьютерный практикум по научному письму > %X second.
1 \documentclass[a4paper,12pt]{article} % Classe de document
2 \usepackage[T1]{fontenc} % Encodage de la police
3
4 \begin{document}
5
6 Bonjour le monde !
7
8 Ceci est mon premier document avec \LaTeX.
9
10 % --- Paragraphe 1 ---
11 Voici un premier paragraphe.
12 Il contient plusieurs phrases pour tester le rendu PDF.
13 On peut écrire librement du texte.
14
15 % --- Paragraphe 2 ---
16 Voici un deuxième paragraphe.
17 Regarde : même si je mets beaucoup d'espaces entre les mots,
18 LaTeX les réduit automatiquement à un seul espace.
19
20 % --- Paragraphe 3 ---
21 Avec les espaces insécables (hard spaces) :
22 M.~Dupont habite à Paris~7e arrondissement.
23 On voit que "M." reste collé à "Dupont", et que "Paris 7e" ne peut pas être
24
25 % --- Paragraphe 4 ---
26 Encore un exemple :
27 Le chapitre~1 présente l'introduction.
28 Sans le ~, LaTeX aurait pu couper "chapitre" et "1" sur deux lignes.
29
30 \end{document}
31

```

{ #fig:002 width=100% }

```

\documentclass[a4paper,12pt]{article} % Classe de document
\usepackage[T1]{fontenc} % Encodage de la police

\begin{document}

Bonjour le monde !

Ceci est mon premier document avec \LaTeX.

% --- Paragraphe 1 ---
Voici un premier paragraphe.
Il contient plusieurs phrases pour tester le rendu PDF.
On peut écrire librement du texte.

% --- Paragraphe 2 ---
Voici un deuxième paragraphe.
Regarde : même si je mets beaucoup d'espaces entre les mots,
LaTeX les réduit automatiquement à un seul espace.

% --- Paragraphe 3 ---
Avec les espaces insécables (hard spaces) :
M.~Dupont habite à Paris~7e arrondissement.
On voit que "M." reste collé à "Dupont", et que "Paris 7e" ne peut pas être coupé

```

```

en fin de ligne.

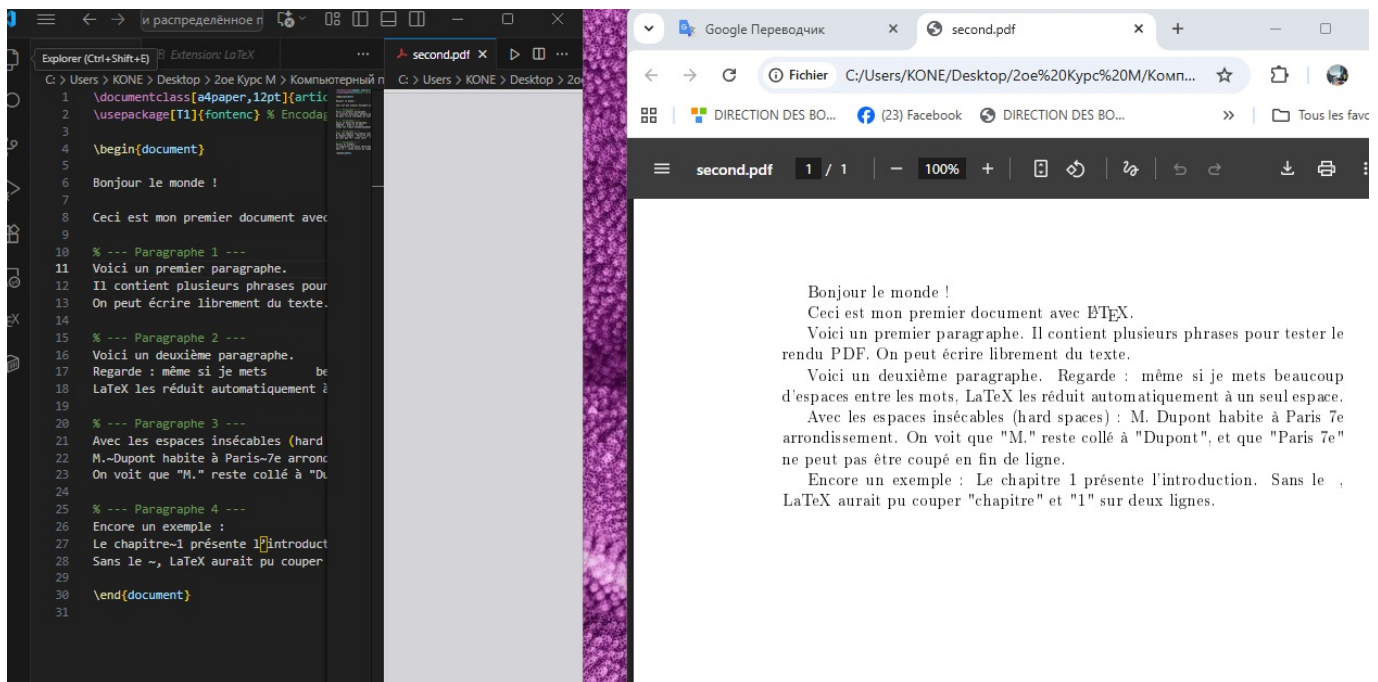
% --- Paragraphe 4 ---
Encore un exemple :
Le chapitre~1 présente l'introduction.
Sans le ~, LaTeX aurait pu couper "chapitre" et "1" sur deux lignes.

\end{document}

```

Make some different paragraphs and add variable spaces.

(см. Рис. [-@fig:010]).



{ #fig:003 width=100% }

Explore how your editor works; click on your source and find how to go to the same line in your PDF.

Try adding some hard spaces and see how they influence line-breaking.

{ #fig:005 width=100% }

(см. Рис. [-@fig:011]).

{ #fig:005 width=100% }

Закключение

Таким образом, была достигнута цель установил TeXlive, я познакомелся с работа с его структуры а также с системой LaTeX.

{.standout}

Спасибо за внимание